NR2DT-06: Step 6 — Migrate Synthetics, SLOs & Workloads

> **Series:** NR2DT | **Notebook:** 6 of 10 | **Created:** April 2026 | **Last Updated:** 04/14/2026

Overview

Goal of this step: complete Wave 2 (synthetics) and Wave 4 (SLOs + workloads). Synthetics are external-facing and visible quickly; SLOs need a 7-day delta-validation window before the math is trusted.

Procedural — see **NRLC-05** (Synthetic Migration) and **NRLC-06** (SLO & Workload Migration) for component depth.

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Prerequisites

Requirement	Details
Audience	Migration lead + assigned engineer for this step
Completed	NR2DT-05 — Migrate Dashboards & Alerts

Requirement	Details
Format	Procedural step — use as a runbook; defer to NRLC for depth
NRLC deep dives	NRLC-05, NRLC-06

1. Wave 2 — Synthetics

Order: HTTP first (lowest risk), then Browser, then API multi-step. Scripted browsers are out of band — they get rebuilt manually.

```
# Inventory + transform
python3 migrate.py --diff --components synthetics

# Import (auto-converts HTTP, Browser, API; stubs scripted)
python3 migrate.py --import-only --components synthetics
```

Re-enter secrets in the DT credentials vault for any monitor that uses bearer tokens, basic auth, or OAuth.

Scripted browser monitors:

1. The transformer emits a stub Browser Monitor with the target URL
2. The original NR JS script is attached as a tag for reference
3. An engineer rebuilds the user journey using DT's clickpath recorder or DSL
4. Validation: confirm the new clickpath catches the same regressions during a 1-week dual-run

G2 — Synthetics SLA: dual-run for ≥ 1 week. DT availability % matches NR within ±0.5%. If delta exceeds, investigate location coverage and credential validity first.

2. Wave 4 — SLOs & Workloads

SLOs depend on metric expressions being mathematically equivalent. Workloads become OpenPipeline enrichments + IAM bucket conditions (Gen3 pattern).

```
python3 migrate.py --diff --components slos,workloads
python3 migrate.py --import-only --components slos,workloads
```

Run the SLO auditor (the most important validation in this wave):

```
python3 migrate.py audit-slos --window 7d --tolerance 0.005
```

Output per SLO:

```
SLO: Checkout Availability
  NR SLI:   99.94%
  DT SLI:   99.92%
  Delta:     0.02%  PASS (within 0.5% tolerance)
```

Any SLO with delta > 0.5% needs investigation:

1. Check unit conversion (ms vs. s vs. ns)
2. Check entity scope — does the bucket / OpenPipeline enrichment filter capture the same entities as NR's workload?
3. Check time alignment — NR's `SINCE 7 days ago` may bin differently than DT's `from:-7d`

G4 — SLO delta: all SLOs report value within tolerance for ≥ 7 days.

3. Step Exit Criteria

G6 — Synthetics + SLOs Migrated

- [ ] All synthetics migrated; G2 dual-run availability ±0.5%
- [ ] Scripted browsers rebuilt and validated
- [ ] All synthetic credentials re-entered in DT vault
- [ ] All SLOs migrated; G4 7-day delta ≤ 0.5%
- [ ] Workload identifiers visible as OpenPipeline-enriched attributes in DQL queries

Next step: NR2DT-07 — Migrate Logs, Tags & Drop Rules.

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